Problem 7.1 (Theoretical + MATLAB)
Falling body system:
Unscented Kalman filter and non-linear system state estimation

On the next page is a system sketch of a body entering the atmosphere from space and falling towards the earth. Also shown is the non-linear continuous time state space model of the system. The state vector consists of the position, speed and ballistic coefficient of the falling body. In this problem will be compared an Extended Kalman filter (EKF) with an Unscented Kalman filter (UKF) in terms of their capabilities to estimate the state vector as the body falls towards earth.

Theoretical part of problem:

Convert the continuous-time non-linear state space model to a discrete-time non-linear state space model on the form

$$\mathbf{x}_k = \mathbf{f}(\mathbf{x}_{k-1}) + \mathbf{w}_{k-1}, \qquad E[\mathbf{w}_k] = 0, \quad Cov[\mathbf{w}_k] = \mathbf{Q}$$
$$z_k = h(\mathbf{x}_k) + v_k, \qquad E[v_k] = 0, \quad E[v_k^2] = \sigma_v^2$$

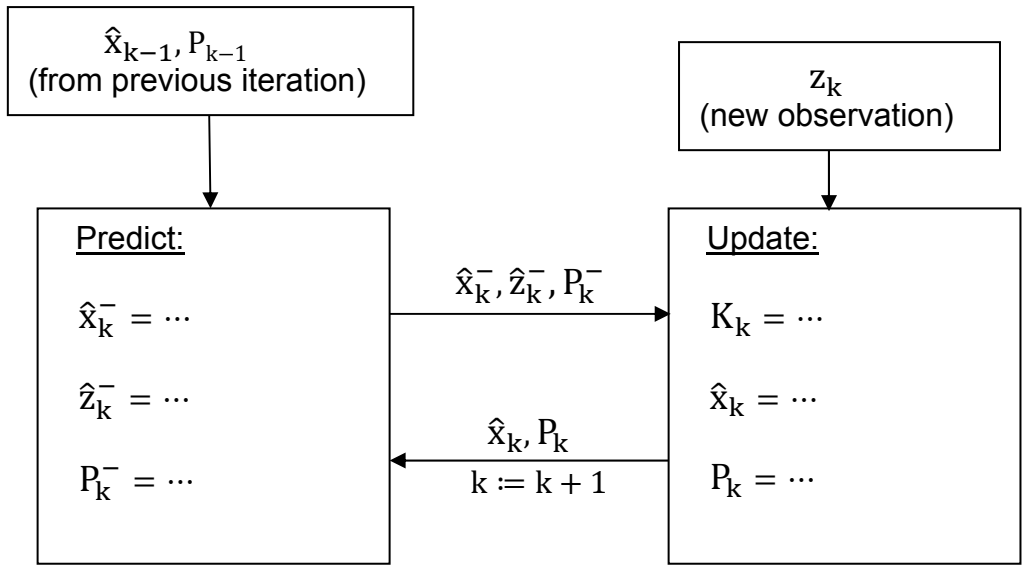
(Hint: use a simple first-order discrete-time approximation to continuous-time derivative:

$\frac{dr}{dt} = s(t) \Rightarrow r_k \cong r_{k-1} + s_{k-1}T_s$ where T_s is the sampling period)

Calculate the Jacobians

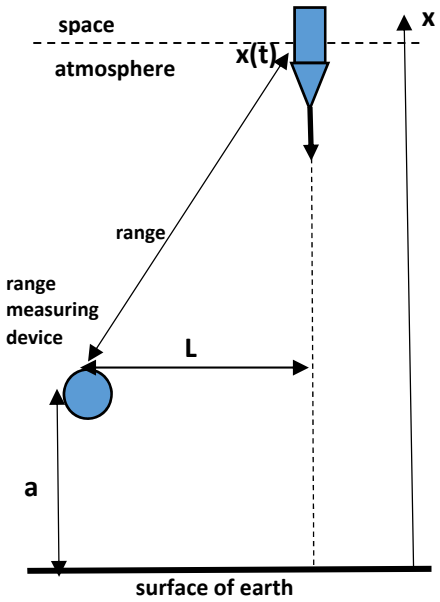
- $\mathbf{A}_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k-1}}$
- $\mathbf{W}_k = \left.\frac{\partial f}{\partial w}\right|_{\hat{x}_{k-1}}$
- $\mathbf{H}_k = \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-}$
- $V_k = \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^-}$

Thus, **find the Extended Kalman filter equations** for this system on the form



Example of non-linear system: Body entering from space through atmosphere and falling towards earth

System



Non-linear system dynamics

$x_1(t) =$ altitude of falling body
$x_2(t) =$ speed of falling body

$x_3(t) =$ ballistic coefficient

Kinematics:

$$\frac{dx_1}{dt} = x_2(t)$$

Newtons 2' law:

$\frac{dx_2}{dt} =$ "drag – gravity"

$$= \frac{1}{2}\rho_0 \, e^{-\frac{x_1(t)}{D}} x_2^2(t) \, x_3(t) - g$$

where

$\rho_0 =$ air density at sea level

$e^{-\frac{x_1(t)}{D}} =$ air density increases exponentially with increasing altitude

$x_2^2(t) =$ drag force increases quadratically with increasing speed

$x_3(t) =$ drag depends on ballistic coefficient

$g =$ gravitational acceleration

Range measurement:

$$z(t) = \sqrt{L^2 + (x_1(t) - a)^2}$$

Non-linear state space model

State: $\mathbf{x}(t) = [x_1(t) \quad x_2(t) \quad x_3(t)]^T$

Observed range: $z(t)$

Process noise: $\mathbf{w}(t) \sim N_3(0, \mathbf{Q})$, $\mathbf{Q} = \begin{bmatrix} \sigma_{w1}^2 & 0 & 0 \\ 0 & \sigma_{w2}^2 & 0 \\ 0 & 0 & \sigma_{w3}^2 \end{bmatrix}$

Range measurement noise: $v(t) \sim N(0, \sigma_v^2)$

Non-linear state space model:

$$\frac{dx_1}{dt} = x_2(t) + w_1(t)$$

$$\frac{dx_2}{dt} = \frac{1}{2}\rho_0 \, e^{-\frac{x_1(t)}{D}} x_2^2(t) \, x_3(t) - g + w_2(t)$$

$$\frac{dx_3}{dt} = w_3(t)$$

$$z(t) = \sqrt{L^2 + (x_1(t) - a)^2} + v(t)$$

MATLAB part of problem:

Implement the non-linear system and an Extended Kalman filter for estimating position, speed and ballistic coefficient of the falling body as a function of time.

Implement also the Unscented Kalman filter algorithm using the Unscented Transform on the non-linearities f and h (define sigma-points, propagate through the non-linearity, define weights, make a priori estimates) combined with the Kalman filter update equations (refer, e.g. to the UKF algorithm slide made available)
For the UT you can, e.g. use $n = 3,\ \kappa = 0,\ n + \kappa = 3$

You can, e.g., use the following system parameters

$N = 80$ samples
$f_s = 2$ Hz
$\rho_0 = 1.225$ kg/m^3
$D = 6$ km
$L = 30$ km
$a = 30$ km
$g = 9.82$ m/s^2

$$\mathbf{Q} = \begin{bmatrix} \sigma_{w1}^2 & 0 & 0 \\ 0 & \sigma_{w2}^2 & 0 \\ 0 & 0 & \sigma_{w3}^2 \end{bmatrix} = \mathbf{0}$$

$\sigma_v^2 = 1100$ m^2

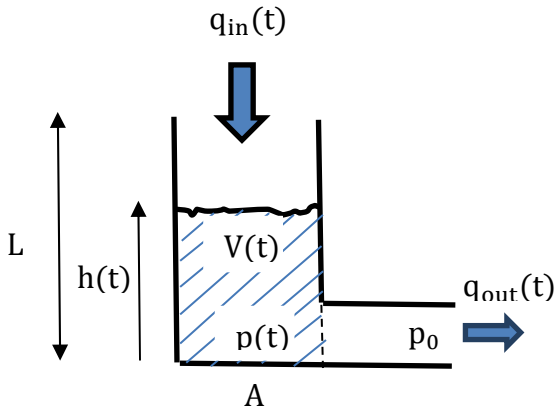
$$\mathbf{x}(0) = [100\text{ km} \quad -6000\text{ m/s} \quad 0.002\text{ m}^2/\text{kg}]^T$$

$$\mathbf{P}_0 = \begin{bmatrix} 100000\text{ m}^2 & 0 & 0 \\ 0 & 500000\ (\text{m/s})^2 & 0 \\ 0 & 0 & 100\ (\text{m}^2/\text{kg})^2 \end{bmatrix}$$

Make plots of true, observed and estimated position, speed and ballistic coefficient using UKF and EKF, respectively.
Also, plot state estimation errors and compare EKF with UKF.

Problem 7.2 (Theoretical + MATLAB)
Water tank system: (continuation of EKF Problem 6.2)
Unscented Kalman filter and non-linear system state estimation

Below is a system sketch of at water tank of height L and bottom area A. The tank that has an input flow $q_{in}(t)$ (control input) and an output flow $q_{out}(t)$. The output flow is determined by the difference between the liquid pressure at the bottom of the tank and the liquid pressure in the outflow channel.



This is the same non-linear system that is considered in Problem 6.2 using the Extended Kalman filter. In this continuation of Problem 6.2, the system will be made even "more non-linear" by making not only the governing differential equation but now also the observation model non-linear (see red colour observation model below).
The purpose is to study the Unscented Kalman filter (Sigma Point KF) by comparing its estimation capabilities with the Extended Kalman filter for this highly non-linear system.

As explained in Problem 6.2 the system dynamics can be put into **continuous-time non-linear state space form:**

true state: $x(t) = h(t)$
observed state: $z(t)$
input: $u(t) = q_{in}(t)$
process noise: $w(t) \sim N(0, \sigma_w^2)$
obs. noise: $v(t) \sim N(0, \sigma_v^2)$

process model: $\frac{dx}{dt} = -\frac{K_v}{A}\sqrt{\rho g x(t) - p_0} + \frac{u(t)}{A} + w(t)$
observation model: $\mathbf{z(t) = x^2(t) + x^3(t) + v(t)}$

Note the changed observation model – now it is also non-linear contrary to Problem 6.2

Theoretical part of problem
(Note that most, but not all, have been done in Problem 6.2):

Convert the continuous-time non-linear state space model to a discrete-time non-linear state space model on the form

$$x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}, \qquad E[w_k] = 0, \quad E\left[w_k^2\right] = \sigma_w^2$$
$$z_k = h(x_k) + v_k, \qquad E[v_k] = 0, \quad E\left[v_k^2\right] = \sigma_v^2$$ **(new for this Problem)**

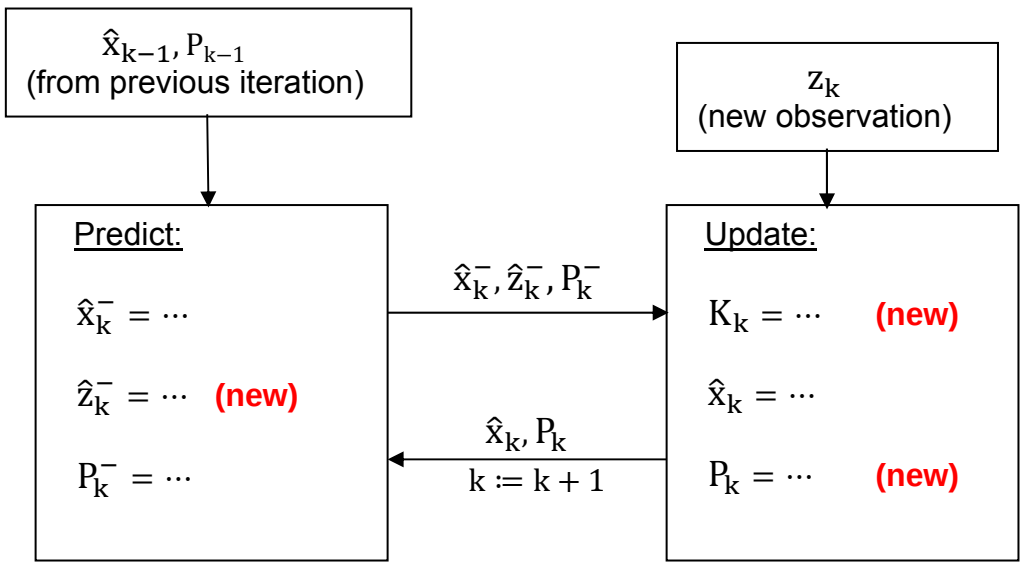
(Hint: use a simple first-order discrete-time approximation to continuous-time derivative:

$\frac{dr}{dt} = s(t) \Rightarrow r_k \cong r_{k-1} + s_{k-1} T_s$ where T_s is the sampling period)

Calculate the Jacobians

- $A_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k-1}}$
- $W_k = \left.\frac{\partial f}{\partial w}\right|_{\hat{x}_{k-1}}$
- $H_k = \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-}$ **(new for this Problem)**
- $V_k = \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^-}$

Thus, **find the Extended Kalman filter equations** for this system on the form



MATLAB part of problem:

Implement the necessary changes from Problem 6.2 for the non-linear system and an Extended Kalman filter for estimating water level as a function of time

Implement also the Unscented Kalman filter algorithm using the Unscented Transform on the non-linearities f and h (define sigma-points, propagate through the non-linearity, define weights, make a priori estimates) combined with the Kalman filter update equations (refer, e.g. to the UKF algorithm slide made available)
For the UT you can, e.g. use $n = 1,\ \kappa = 2,\ n + \kappa = 3$

You can, e.g., use the following system parameters (given without units)

$N = 100$
$f_s = 10$
$L = 1 \ \Rightarrow \ 0 \leq h(t) \leq 1$
$A = 1$
$\rho = 1000$
$g = 9.82$
$p_0 = 1000$
$K_v = 0.01$
$\sigma_w^2 = 10^{-4}$
$\sigma_v^2 = 10^{-2}$
$q_{in}(t) = q_{in} = 0.5$

Make plots of true, observed and estimated water levels using UKF and EKF
Plot estimation errors and calculate MSE's

Problem 7.3 (MATLAB)
Unscented Transform:
Propagation of mean and covariance through non-linearity

In this problem will be compared the Extended Kalman filter (EKF) approach for propagating random variables and their mean and covariance through a non-linear function, that is linearization, with the Unscented Kalman filter approach, which is defining so called sigma points and using the Unscented Transform (UT).

As an example, consider conversion of polar to rectangular coordinates in 2D as an example of a non-linear transform. That is consider a vector random variable $\mathbf{x} = [x_1\ x_2] = [r\ \theta]$ transformed to another vector random variable $\mathbf{y} = \mathbf{f}(\mathbf{x}) = [y_1\ y_2] = [r*\cos(\theta)\ \ r*\sin(\theta)]$.

- Define 500 random vectors of **x**, where r is uniformly distributed between 0,99 and 1.01 and θ is uniformly distributed between π/2-0.35 and π/2+-0.35 (and independent of r)
- Plot the points and their mean in an (r, θ) coordinate system.

- Calculate and plot the transformed **y** vectors in an $(y_1\ y_2)$ coordinate system
- Plot also the true mean of the **y** vectors and the linearized mean, that is the transformed mean of the **x** vectors and note the difference (this is how EKF works).

- Calculate also the linearized covariance matrix, which is $\text{grad}(\mathbf{f})*\text{Cov}(x)*\text{grad}(\mathbf{f})^T$ (this is what EKF does) and the true (empirical) covariance matrix of **y**
- Visualize the covariance matrices as ellipses where the axes half lengths are the square root of the eigenvalues of the covariance matrix and the axes directions are given by the eigenvectors
- Note the differences

- Next, turning to the Unscented Transform approach

- Define and plot the **x** sigma points
- Propagate them through the non-linearity to **y** sigma points
- Plot the **y** sigma points and their mean together with the true mean and the linearized mean (from above)
- Comment

- Calculate and visualize also the covariance matrix of the **y** sigma points in the same way as above and compare with the two covariance matrix plots from above.
- Comment

- Make an overall conclusion with respect to the capabilities of the EKF and UT approaches for propagating mean and covariance through a non-linear transformation